Scientific revision summary

Verdict: B — Near ready as a focused theory paper. Thirteen of 19 tracked concerns are resolved, one is partially resolved and five remain unresolved. All identified P0/P1 review items have concrete scientific or manuscript remedies. The remaining four general theory problems and real-data extension are not marked solved. The complete response matrix, technical dependency audit and second adversarial review accompany this summary.

A. Complete review registry

The registry was created after reading the full review and before scientific edits. It preserves F01–F10, adds the cross-cutting novelty, precision, data and input-cost concerns, and separately tracks T01–T05. See issue registry, full machine-readable matrix and individual responses. The 40-item crosswalk records every instruction in the revision protocol.

B. Critical fixes

F01: coverage evaluation. A target-scaled tolerance counted a genuine 90% miss. The new policy permits only four representable endpoint steps, retains strict counts and distinguishes evaluation rounding from statistical slack. All 303 saved interval-producing cases were reevaluated. The two affected XCH/oracle rows in S03_007 change from 134/150 to 133/150; they share one underlying observation. All original endpoints remain unchanged. The earlier exact-noise replay and the new scale checks support the correction.

F03: one current submission. The revised main paper presents final all-budget rates, the tensor-specific mechanism and current limitations. A complete technical supplement contains all retained proofs and R1. A current empirical report contains corrected saved results and new experiments. The reading order and status ledger distinguish this revision from all preserved earlier drafts.

C. Major fixes

- **F02:** controlled component comparisons replace the confounded causal interpretation. Pure bias removal has no effect when fallback is selected; pure fallback removal retains intersections. The original combined diagnostic remains visible under an accurate name.
- **F04:** R1 introduces a separate scalar companion with a complete risk proof. The saved S03_002 MSE falls from 12,084,661.379 to 0.147776 on the same 500 observations, with unchanged intervals. Old rare-event losses remain available. This does not establish P1 minimax squared risk.
- **F05:** a ten-level fixed-M sweep separates increasing signal from a changing supplied class, with a paired norm-comparable branch.
- **F06:** the XCH method is mapped to its tensor-PCA source, and case applicability is separate from guarantee type. Unknown regularity constants remain explicitly unverified; shorter regular-signal comparator intervals are retained.
- **F07:** the ideal residual optimizer, binary float64 search and executable conservative methods are explicitly distinguished. A fixed-loading study has 7/600 endpoint certificates and retains 593 unresolved solves.
- **N01:** the paper's novelty rests on the full tensor likelihood, cubic harmonic lower bound and ordered all-budget residual argument. Classical sparse-functional and honest-adaptation contributions are credited.

D. Theory changes

No central theorem rate or original assumption was silently changed. The status ledger contains **59 retained PROVED results plus one new PROVED proposition, R1**. These are complete internally checked supplied proofs, not formal verification or external peer review.

R1 assumes a direct Gaussian scalar observation with valid structural range and uses the existing direct/structural intersection. It bounds unconditional squared risk by C_alpha min(h^2,tau^2), without pilot-success conditioning, separation or candidate independence. At h tau it uses zero; otherwise it uses the intersection midpoint, with clipped direct estimation for an empty/zero sentinel. The complete pointwise tau^2 r/p risk claim remains open.

The assumption audit explains which arguments survive joint sub-Gaussian errors, finite-variance independent replicates, known row corruption and independent Gaussian residual scale. It also identifies where arbitrary dependence, unknown directional calibration, selected loadings and underestimated bounds fail. The independent-scale

and row-corruption theorems were already available; implementing and testing them is an empirical revision, not a new robustness theorem.

E. New simulations and reevaluation

There are **87,200 new independent datasets in 55 registered configurations**: 10,000 fixed-budget/ablation draws; 200 binary-model observations shared across three budgets, giving 600 solves; 19,000 row-corruption/noise draws; 54,000 variance-calibration draws; and 4,000 larger pointwise draws. The new summaries contain 612 method/level rows.

The original **1,084,640 saved datasets are preserved**, not recounted as new. All 303 interval cases were reevaluated under the new event/estimator definitions. The 15 remaining stochastic cases measure calibration events and have no affected intervals. The original study's 84 analytic-only configurations are still labeled analytic. All ten saved-study figures were regenerated; six new vector figures display the final rates and revision experiments. Numerical main-text statements and tables are generated from checked outputs.

Negative findings remain: conservative calibrated intervals, weak-signal XCH failure outside certified regularity, severe corruption-cap failure, raw catastrophic midpoint losses, and 593 unresolved optimizer searches. Heavy-tail MSE sample diagnostics are not used as justified fourth-moment error bars. Larger pointwise results are reported with Wilson intervals and rate-normalized lengths rather than an invented exact asymptotic slope.

F. Real-data changes

No defensible repeated-tensor dataset, independent sampling unit, prescribed contrast and noise/bound calibration were established from the supplied materials. No real-data or LLM inference result was fabricated. The focused manuscript remains theoretical with simulations. The broader application request remains unresolved and requires a scientifically justified dataset/model design.

G. Manuscript and artifact changes

Read main paper, complete technical supplement and empirical supplement. The main text follows the final logical argument; preliminary bounds and full proofs are in the supplement. The abstract was generated after the scientific fixes and new results stabilized. The notation dictionary separates dimension, degree, diagonal-envelope coefficients, approximation error and pilot bounds. The current API returns interval and scalar estimate separately.

The original project, review, simulations and prior uncommitted work are preserved. The new stage is isolated in review_revision; the root README and requirements ledger are updated to point to the final state. No commit, push, external message or submission was made.

H. Remaining limitations

Remaining issue	Classification	Status
General efficient fixed-cap diffuse weak-signal minimax computation	FUTURE WORK; SUBMISSION RISK if advertised as a computational method	NOT RESOLVED
Intermediate growing-cap sharp rates	FUTURE WORK	NOT RESOLVED
Arbitrary loading profiles beyond matched regimes	FUTURE WORK	NOT RESOLVED
Complete pointwise/TT adaptation and optimal SNR	FUTURE WORK	NOT RESOLVED
Broader unknown-scale/dependent/selected-loading calibration	FUTURE WORK	PARTIALLY RESOLVED through available special-case calibration only
A valid real-data application	ACCEPTABLE LIMITATION for a theory paper; unresolved broader task	NOT RESOLVED

Remaining issue	Classification	Status
Breadth and external novelty priority	SUBMISSION RISK	Requires specialist/latest-version assessment
Conservative constants, finite Monte Carlo precision and numerical arithmetic	ACCEPTABLE LIMITATION	Explicitly measured and reported

No known blocking correctness issue remains for the focused claims. This is not a statement that a complete efficient arbitrary-loading adaptive theory has been achieved.

I. Revised mathematical contribution

For a normalized flat orthogonal contrast in the full Gaussian tensor mean experiment, fixed order K 3, positive fixed Tucker or TT caps, known tau and supplied M, write ell=log(ep) and x=M/(tau sqrt(ell)). Honest expected length at zero and worst case both have order

$$\tau\sqrt{\ell/p}\,\min\{x, 1+\log_+ x, \ell\} \quad (K=3),$$

and $\tau\sqrt{\ell/p}\,\min\{x,1\}$ for K 4; minimax squared risk is the square of the length rate. The tensor-specific content is the full-data residual and harmonic-mixture argument. Special growing-cap/loading results and adaptation boundaries broaden this claim within explicit limits. R1 repairs the estimator output of an executable interval while asserting only its proved noise-scale risk.

J. Journal recommendation

Bernoulli is the best current target in my assessment, provided a specialist confirms the novelty and significance of the full-data cubic/all-budget argument. Its stated emphasis on theoretical mathematical statistics is consistent with this focused formulation. This is a fit judgment, not an acceptance prediction. Bernoulli aims and scope.

Electronic Journal of Statistics is a credible alternative for the combination of detailed theory, explicit computational scope and reproducible simulations; it publishes theoretical, computational and applied statistics. IMS description of EJS.

The other requested venue assessments are editorial judgments about this manuscript rather than statements of journal policy:

Venue	Current fit assessment
Annals of Statistics	Possible but ambitious; breadth and priority of the mathematical mechanism would need especially persuasive specialist support. The long auxiliary theory should not obscure the central advance.
JASA — Theory and Methods	Greater risk from lack of a general implementable optimal method or a defensible application. A stronger computational result would materially improve fit.
JRSS-B	Similar concern about broadly usable methodology and scientific relevance; current theory-led framing is narrower.
Biometrika	A sharply distilled main theorem and mechanism could fit a concise theoretical contribution, but the broad multi-extension narrative should remain in supplements.
Bernoulli	Strongest match to the present specialized theoretical emphasis, conditional on novelty/significance.

Venue	Current fit assessment
EJS	Natural alternative with room for technical detail and transparent reproducibility.

Scientific before → after

Component	Before revision	After revision
Coverage evaluation	Target-scaled tolerance counted one real miss twice across paired methods	Bounded-ULP policy, strict counts, all affected observations reevaluated
Scalar estimation	Midpoint could incur order-M^2 sentinel loss	Separate R1 estimator with unconditional noise-scale bound; raw losses retained
Pointwise theorem	Expected length only	Same expected-length theorem; no unsupported MSE upgrade
Ablation	Bias, fallback/intersection and allocation changed together	Paired isolated interventions and accurately named combined diagnostic
Signal sweep	Signal amplitude and supplied M changed together	Fixed-M sweep plus paired norm-comparable branch
Computational benchmark	Loading/tolerance changes confounded budget comparisons	Same loading, direction, class and tolerance; all 600 gaps recorded
Robust refinements	Row-corruption and residual-scale propositions lacked direct tests	73,000 new datasets testing those propositions and explicit violations
Large-p precision	150–300 draws at largest dimensions	1,000 new independent draws in each larger case; Wilson uncertainty
Submission structure	Chronological main draft plus later supplements	One focused main paper, complete proof supplement and current empirical report
Theory scope	Four broad open problems remained	Still open; now separated from resolved review defects and the new R1 proposition

The second adversarial review contains the full readiness checkpoint and strongest remaining skeptical objection. The validation log records newly run checks separately from preserved prior evidence.